

Executive Summary

- **Objective:**

The analysis aims to identify the major factors influencing **customer churn**, with emphasis on customer demographics, tenure, contract type, payment methods, charges, and subscribed services.

- **Churn Distribution:**

Customer churn is analyzed by comparing customers who **stayed with the company** against those who **discontinued their services**. This provides the overall picture of customer retention and helps identify segments with a higher risk of churn.

- **Customer Tenure:**

Customers with **shorter tenure** tend to be more vulnerable to churn, indicating that the early stage of the customer relationship is critical for retention. Long-term customers generally demonstrate stronger retention.

- **Contract Type:**

Month-to-month customers show a higher tendency to churn than customers with one-year or two-year contracts. Longer-term contracts can therefore be associated with greater customer stability.

- **Payment Method:**

Electronic check customers represent an important segment in churn analysis and show higher churn behavior compared with several other payment methods. Payment behavior should therefore be considered when developing retention strategies.

- **Monthly Charges:**

Customers with **higher monthly charges** can represent a higher-risk segment. Combining MonthlyCharges with tenure and contract type can provide better insight into customers who may be considering cancellation.

- **Internet Service:**

Fiber optic is the largest internet-service category at approximately **44.0%**, followed by **DSL at 34.4%**. Approximately **21.7%** of

customers have no internet service. Comparing these categories with Churn can help identify differences in retention.

- **Phone Service:**

PhoneService has very high adoption, with approximately **90.3%** of customers using the service. Because it is used by most customers, its relationship with churn should be examined together with other variables rather than independently.

- **Additional Services:**

Adoption of optional services is comparatively lower. **OnlineSecurity is used by approximately 28.7%** of customers and **TechSupport by approximately 29.0%**. OnlineBackup and DeviceProtection have adoption rates of approximately **34.5%** and **34.4%**, respectively.

- **Streaming Services:**

Approximately **38.4%** of customers use StreamingTV and **38.8%** use StreamingMovies. These services provide useful customer-segmentation variables for comparing churn patterns.

- **Overall Conclusion:**

The analysis suggests that **Churn is influenced by a combination of customer tenure, contract type, payment method, monthly charges, internet service, and service subscriptions**. Rather than considering a single variable, these factors should be analyzed together to identify high-risk customer segments and develop targeted retention strategies.